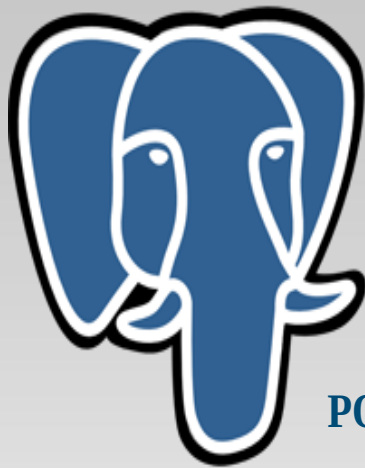




PostgreSQL

POSTGRES DATABASE ADMINISTRATION



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[SENIOR DATABASE ADMINISTRATOR|ORACLE| POSTGRESQL | MSSQL|OCI|AWS]

CHAPTER 3: POSTGRESQL PHYSICAL STRUCTURES

About the Author:

I am **Stephen Njoroge**, a Senior Database Engineer with over 8+ years of expertise in designing, securing and optimizing high-performance database systems across the banking and fintech sectors. My core technical proficiency spans major databases: Oracle, PostgreSQL, MS SQL Server and MySQL-along with advanced scripting in SQL and Python for automation and data pipeline development. I architect and manage cloud and virtualization solutions on OCI, AWS, and Proxmox for onprem systems and implement comprehensive DevOps practices using monitoring tools like Prometheus and Grafana and IT service management via iTop. Further, I deliver advanced data analytics and visualization using Tableau, Power BI and Oracle BI. As a Certified Data Protection Officer, I integrate robust data protection, compliance and security measures including vulnerability assessment and role-based access control into every layer of the data infrastructure.

Chapter 3: PostgreSQL Physical Structures

PostgreSQL Database Cluster.

Meaning:

A Database Cluster in PostgreSQL is a single instance of PostgreSQL server managing multiple databases under one filesystem location, all running on the same physical or virtual machine.

Showing the existing clusters:

- Using pg_lsclusters

```
# List all clusters
pg_lsclusters
```

```
steve@998-986-02:~$ pg_lsclusters
Ver Cluster Port Status Owner    Data directory          Log file
18  main     5432  online postgres /var/lib/postgresql/18/main log/postgresql-%Y-%m-%d_%H%M%S.csv.csv
steve@998-986-02:~$
```

- Check Running Processes

```
# Look for PostgreSQL processes
ps aux | grep postgres
```

```
steve@998-986-02:~$ ps aux | grep postgres
postgres    354  0.0  0.9 230960 36316 ?        Ss   17:01   0:00 /usr/lib/postgresql/18/bin/postgres -D /var/lib/postgresql/18/main
-c config_file=/etc/postgresql/18/main/postgresql.conf
postgres    379  0.0  0.1 75264 5156 ?        Ss   17:01   0:00 postgres: 18/main: logger
postgres    396  0.0  0.1 230960 7828 ?        Ss   17:01   0:00 postgres: 18/main: io worker 0
postgres    397  0.0  0.1 230960 6704 ?        Ss   17:01   0:00 postgres: 18/main: io worker 1
postgres    398  0.0  0.1 230960 5796 ?        Ss   17:01   0:00 postgres: 18/main: io worker 2
postgres    399  0.0  0.2 231084 8940 ?        Ss   17:01   0:00 postgres: 18/main: checkpointer
postgres    400  0.0  0.1 230960 5860 ?        Ss   17:01   0:00 postgres: 18/main: background writer
postgres    513  0.0  0.2 230960 10396 ?       Ss   17:01   0:00 postgres: 18/main: walwriter
postgres    514  0.0  0.2 232520 8384 ?        Ss   17:01   0:00 postgres: 18/main: autovacuum launcher
postgres    515  0.0  0.1 230960 6956 ?        Ss   17:01   0:00 postgres: 18/main: archiver failed on 000000010000000000000001
postgres    516  0.0  0.1 232388 6636 ?        Ss   17:01   0:00 postgres: 18/main: logical replication launcher
steve       820  0.0  0.0 4096 1988 pts/0    S+   17:13   0:00 grep --color=auto postgres
steve@998-986-02:~$ |
```

```
# more specifically
ps aux | grep "postgres:"
```

```

steve@998-986-02:~$ ps aux | grep "postgres:"
postgres 379 0.0 0.1 75264 5156 ? Ss 17:01 0:00 postgres: 18/main: logger
postgres 396 0.0 0.1 230960 7828 ? Ss 17:01 0:00 postgres: 18/main: io worker 0
postgres 397 0.0 0.1 230960 6704 ? Ss 17:01 0:00 postgres: 18/main: io worker 1
postgres 398 0.0 0.1 230960 5796 ? Ss 17:01 0:00 postgres: 18/main: io worker 2
postgres 399 0.0 0.2 231084 8940 ? Ss 17:01 0:00 postgres: 18/main: checkpointer
postgres 400 0.0 0.1 230960 5860 ? Ss 17:01 0:00 postgres: 18/main: background writer
postgres 513 0.0 0.2 230960 10396 ? Ss 17:01 0:00 postgres: 18/main: walwriter
postgres 514 0.0 0.2 232520 8384 ? Ss 17:01 0:00 postgres: 18/main: autovacuum launcher
postgres 515 0.0 0.1 230960 6956 ? Ss 17:01 0:00 postgres: 18/main: archiver failed on 000000010000000000000001
postgres 516 0.0 0.1 232388 6636 ? Ss 17:01 0:00 postgres: 18/main: logical replication launcher
steve 1072 0.0 0.0 4096 1984 pts/0 S+ 17:13 0:00 grep --color=auto postgres:
steve@998-986-02:~$ |

```

- Check Service Status (Systemd)

```

# List PostgreSQL services
systemctl list-units | grep postgresql

```

```

steve@998-986-02:~$ systemctl list-units | grep postgresql
postgresql.service                                loaded active   exited        PostgreSQL RDBMS
postgresql@18-main.service                        loaded active   running        PostgreSQL Cluster 18-main
system-postgresql.slice                           loaded active   active         Slice /system/postgresql
steve@998-986-02:~$

```

```

# Checking status of a specific version
systemctl status postgresql@18-main

```

```

steve@998-986-02:~$ systemctl status postgresql@18-main
● postgresql@18-main.service - PostgreSQL Cluster 18-main
   Loaded: loaded (/usr/lib/systemd/system/postgresql@.service; enabled-runtime; preset: enabled)
   Active: active (running) since Tue 2026-01-13 17:00:13 MSK; 17min ago
     Process: 236 ExecStart=/usr/bin/pg_ctlcluster --skip-systemctl-redirect 18-main start (code=exited, status=0/SUCCESS)
    Main PID: 354 (postgres)
      Tasks: 11 (limit: 4597)
     Memory: 45.4M ()
    CGroup: /system.slice/system-postgresql.slice/postgresql@18-main.service
            └─354 /usr/lib/postgresql/18/bin/postgres -D /var/lib/postgresql/18/main -c config_file=/etc/postgresql/18/main/postgre>
              └─379 "postgres: 18/main: logger "
                └─396 "postgres: 18/main: io worker 0"
                  └─397 "postgres: 18/main: io worker 1"
                    └─398 "postgres: 18/main: io worker 2"
                      └─399 "postgres: 18/main: checkpointer "
                        └─400 "postgres: 18/main: background writer "
                          └─513 "postgres: 18/main: walwriter "
                            └─514 "postgres: 18/main: autovacuum launcher "
                              └─515 "postgres: 18/main: archiver failed on 000000010000000000000001"
                                └─516 "postgres: 18/main: logical replication launcher "

```

```

Jan 13 17:00:05 998-986-02 systemd[1]: Starting postgresql@18-main.service - PostgreSQL Cluster 18-main...
Jan 13 17:00:06 998-986-02 postgresql@18-main[236]: Removed stale pid file.
Jan 13 17:00:13 998-986-02 systemd[1]: Started postgresql@18-main.service - PostgreSQL Cluster 18-main.

```

- Check Data Directories

```

# Common default postgresql locations
ls -la /var/lib/postgresql/

```

```

steve@998-986-02:~$ ls -la /var/lib/postgresql/
total 48
drwxr-xr-x  4 postgres postgres 4096 Dec 23 15:56 .
drwxr-xr-x 37 root      root      4096 Dec 21 21:58 ..
-rw-----  1 postgres postgres 1640 Dec 23 17:32 .bash_history
-rw-rw-r--  1 postgres postgres  47 Dec 23 15:56 .bashrc
drwxr-xr-x  2 postgres postgres 4096 Dec 21 19:34 .landscape
-rw-----  1 postgres postgres  20 Dec 14 14:09 .lessht
-rw-rw-r--  1 postgres postgres   0 Dec 23 16:59 .motd_shown
-rw-----  1 postgres postgres 8985 Dec 23 17:32 .psql_history
-rw-----  1 postgres postgres 6500 Dec 23 15:40 .viminfo
drwxr-xr-x  3 postgres postgres 4096 Dec 21 19:13 18
steve@998-986-02:~$

```

```

root@998-986-02:/var/lib/postgresql/18/main# ls -lrt
total 92
drwx----- 2 postgres postgres 4096 Dec 21 19:13 pg_twophase
drwx----- 2 postgres postgres 4096 Dec 21 19:13 pg_snapshots
drwx----- 2 postgres postgres 4096 Dec 21 19:13 pg_serial
drwx----- 2 postgres postgres 4096 Dec 21 19:13 pg_replslot
drwx----- 2 postgres postgres 4096 Dec 21 19:13 pg_notify
drwx----- 4 postgres postgres 4096 Dec 21 19:13 pg_multixact
drwx----- 2 postgres postgres 4096 Dec 21 19:13 pg_dynshmem
drwx----- 2 postgres postgres 4096 Dec 21 19:13 pg_commit_ts
-rw----- 1 postgres postgres   3 Dec 21 19:13 PG_VERSION
drwx----- 2 postgres postgres 4096 Dec 21 19:13 pg_xact
drwx----- 2 postgres postgres 4096 Dec 21 19:13 pg_subtrans
drwx----- 2 postgres postgres 4096 Dec 21 21:16 pg_stat
drwx----- 4 postgres postgres 4096 Dec 23 16:46 pg_wal
drwx----- 9 postgres postgres 4096 Dec 23 16:46 base
drwx----- 2 postgres postgres 4096 Dec 23 17:27 pg_tblspc
-rw----- 1 postgres postgres  117 Dec 23 17:28 postgresql.auto.conf
drwx----- 2 postgres postgres 4096 Jan 13 17:00 pg_stat_tmp
drwx----- 2 postgres postgres 4096 Jan 13 17:00 log
-rw----- 1 postgres postgres   48 Jan 13 17:00 current_logfiles
-rw----- 1 postgres postgres  130 Jan 13 17:00 postmaster.opts
-rw----- 1 postgres postgres  107 Jan 13 17:00 postmaster.pid
drwx----- 2 postgres postgres 4096 Jan 13 17:02 global
drwx----- 4 postgres postgres 4096 Jan 13 17:05 pg_logical
root@998-986-02:/var/lib/postgresql/18/main# pwd
/var/lib/postgresql/18/main
root@998-986-02:/var/lib/postgresql/18/main# |

```

```

# Check PGDATA environment variable
echo $PGDATA

```

```

root@998-986-02:/var/lib/postgresql/18/main# echo $PGDATA
/var/lib/postgresql/18/main
root@998-986-02:/var/lib/postgresql/18/main# |

```

- Check Configuration Files

```
# Find postgresql.conf files
find / -name "postgresql.conf" 2>/dev/null
```

```
root@998-986-02:/var/lib/postgresql/18/main# find / -name "postgresql.conf" 2>/dev/null
/etc/postgresql/18/main/postgresql.conf
/mnt/c/pgbackup/base/postgresql.conf
/mnt/c/Program Files/PostgreSQL/15/data/postgresql.conf
/mnt/c/Program Files/PostgreSQL/17/data/postgresql.conf
/mnt/c/Program Files/PostgreSQL/18/data/postgresql.conf
/mnt/c/Program Files/PostgreSQL/18/data_old/postgresql.conf
```

```
# Check default config location
ls -la /etc/postgresql/
```

```
root@998-986-02:/var/lib/postgresql/18/main# ls -la /etc/postgresql/
total 12
drwxr-xr-x  3 postgres postgres 4096 Dec 22 16:16 .
drwxr-xr-x 95 root      root      4096 Jan 13 17:00 ..
drwxr-xr-x  3 postgres postgres 4096 Dec 21 19:13 18
root@998-986-02:/var/lib/postgresql/18/main#
```

- From Inside PostgreSQL

```
-- Show cluster information
```

```
SHOW data_directory;
SHOW config_file;
SHOW hba_file;
```

```
postgres=# SHOW data_directory;
SHOW config_file;
SHOW hba_file;
      data_directory
-----
/var/lib/postgresql/18/main
(1 row)

      config_file
-----
/etc/postgresql/18/main/postgresql.conf
(1 row)

      hba_file
-----
/etc/postgresql/18/main/pg_hba.conf
(1 row)

postgres=# |
```

-- Version and uptime

```
SELECT version();
SELECT pg_postmaster_start_time();
```

```
postgres=# SELECT version();
               version
-----
PostgreSQL 18.0 on x86_64-windows, compiled by msvc-19.44.35215, 64-bit
(1 row)
```

```
postgres=# SELECT pg_postmaster_start_time();
 pg_postmaster_start_time
-----
2026-01-06 09:12:47.944177+03
(1 row)
```

```
postgres=# |
```

- Using pg_ctl

Check status of a specific data directory

```
pg_ctl -D /var/lib/postgresql/18/main status
```

```
postgres@998-986-02:/var/lib$ pg_ctl -D /var/lib/postgresql/18/main status
pg_ctl: server is running (PID: 354)
/usr/lib/postgresql/18/bin/postgres "-D" "/var/lib/postgresql/18/main" "-c" "config_file=/etc/postgresql/18/main/postgresql.conf"
postgres@998-986-02:/var/lib$ |
```

- Check Multiple Ports

PostgreSQL can run on different ports

```
netstat -tulpn | grep LISTEN | grep -E "543[0-9]"
```

```
root@998-986-02:/var/lib/postgresql/18# netstat -tulpn | grep LISTEN | grep -E "543[0-9]"
tcp        0      0 127.0.0.1:5432          0.0.0.0:*               LISTEN      354/postgres
root@998-986-02:/var/lib/postgresql/18#
```

or

```
lsof -i -P -n | grep LISTEN | grep postgres
```

```
root@998-986-02:/var/lib/postgresql/18# lsof -i -P -n | grep LISTEN | grep postgres
postgres 354      postgres 7u IPv4 309      0t0  TCP 127.0.0.1:5432 (LISTEN)
root@998-986-02:/var/lib/postgresql/18#
```

Creating a cluster:

Here are several methods to create a new PostgreSQL cluster:

- Check if PostgreSQL is installed:

```
postgres --version
initdb --version
```

```
postgres@998-986-02:~/18$ postgres --version
postgres (PostgreSQL) 18.1 (Ubuntu 18.1-1.pgdg24.04+2)
postgres@998-986-02:~/18$ initdb --version
initdb (PostgreSQL) 18.1 (Ubuntu 18.1-1.pgdg24.04+2)
postgres@998-986-02:~/18$
```

- Using initdb

```
# Create a new data directory
mkdir -p /var/lib/postgresql/18/data
```

```
postgres@998-986-02:~/18$ mkdir -p /var/lib/postgresql/18/data
postgres@998-986-02:~/18$ ls
data main
postgres@998-986-02:~/18$
```

```
# Initialize the cluster
initdb -D /var/lib/postgresql/18/data
```

```
postgres@998-986-02:~/18$ initdb -D /var/lib/postgresql/18/data
The files belonging to this database system will be owned by user "postgres".
This user must also own the server process.
```

```
The database cluster will be initialized with locale "C.UTF-8".
The default database encoding has accordingly been set to "UTF8".
The default text search configuration will be set to "english".
```

```
Data page checksums are enabled.
```

```
fixing permissions on existing directory /var/lib/postgresql/18/data ... ok
creating subdirectories ... ok
selecting dynamic shared memory implementation ... posix
selecting default "max_connections" ... 100
selecting default "shared_buffers" ... 128MB
selecting default time zone ... Europe/Volgograd
creating configuration files ... ok
running bootstrap script ... ok
performing post-bootstrap initialization ... ok
syncing data to disk ... ok
```

```
initdb: warning: enabling "trust" authentication for local connections
```

```
initdb: hint: You can change this by editing pg_hba.conf or using the option -A, or --auth-local and --auth-host, the next time you run initdb.
```

```
Success. You can now start the database server using:
```

```
pg_ctl -D /var/lib/postgresql/18/data -l logfile start
```

```
postgres@998-986-02:~/18$ |
```

```
# Initialize with specific locale (recommended)
initdb -D /var/lib/postgresql/18/data --encoding=UTF8
```

```
postgres@998-986-02:~/18/data$ initdb -D /var/lib/postgresql/18/data --encoding=UTF8
The files belonging to this database system will be owned by user "postgres".
This user must also own the server process.
```

The database cluster will be initialized with locale "C.UTF-8".
The default text search configuration will be set to "english".

Data page checksums are enabled.

```
fixing permissions on existing directory /var/lib/postgresql/18/data ... ok
creating subdirectories ... ok
selecting dynamic shared memory implementation ... posix
selecting default "max_connections" ... 100
selecting default "shared_buffers" ... 128MB
selecting default time zone ... Europe/Volgograd
creating configuration files ... ok
running bootstrap script ... ok
performing post-bootstrap initialization ... ok
syncing data to disk ... ok
```

initdb: warning: enabling "trust" authentication for local connections

initdb: hint: You can change this by editing pg_hba.conf or using the option -A, or --auth-local and --auth-host, the next time you run initdb.

Success. You can now start the database server using:

```
pg_ctl -D /var/lib/postgresql/18/data -l logfile start
```

```
postgres@998-986-02:~/18/data$
```

Initialize with custom user

```
initdb -D /var/lib/postgresql/18/data --username=postgres
```

```
postgres@998-986-02:~/18/data$ initdb -D /var/lib/postgresql/18/data --username=postgres
The files belonging to this database system will be owned by user "postgres".
This user must also own the server process.
```

The database cluster will be initialized with locale "C.UTF-8".
The default database encoding has accordingly been set to "UTF8".
The default text search configuration will be set to "english".

Data page checksums are enabled.

```
fixing permissions on existing directory /var/lib/postgresql/18/data ... ok
creating subdirectories ... ok
selecting dynamic shared memory implementation ... posix
selecting default "max_connections" ... 100
selecting default "shared_buffers" ... 128MB
selecting default time zone ... Europe/Volgograd
creating configuration files ... ok
running bootstrap script ... ok
performing post-bootstrap initialization ... ok
syncing data to disk ... ok
```

initdb: warning: enabling "trust" authentication for local connections

initdb: hint: You can change this by editing pg_hba.conf or using the option -A, or --auth-local and --auth-host, the next time you run initdb.

Success. You can now start the database server using:

```
pg_ctl -D /var/lib/postgresql/18/data -l logfile start
```

```
postgres@998-986-02:~/18/data$
```

Full initialization with common options

```
initdb \
-D /var/lib/postgresql/mydata \
--encoding=UTF8 \
--locale=en_US.UTF-8 \
--username=postgres \
--auth=md5 \
--pwprompt
```

```

postgres@998-986-02:~/18/data$ initdb \
-D /var/lib/postgresql/mydata \
--encoding=UTF8 \
--locale=C.UTF-8 \
--username=postgres \
--auth=md5 \
--pwprompt
The files belonging to this database system will be owned by user "postgres".
This user must also own the server process.

The database cluster will be initialized with locale "C.UTF-8".
The default text search configuration will be set to "english".

Data page checksums are enabled.

Enter new superuser password:
Enter it again:

creating directory /var/lib/postgresql/mydata ... ok
creating subdirectories ... ok
selecting dynamic shared memory implementation ... posix
selecting default "max_connections" ... 100
selecting default "shared_buffers" ... 128MB
selecting default time zone ... Europe/Volgograd
creating configuration files ... ok
running bootstrap script ... ok
performing post-bootstrap initialization ... 2026-01-13 17:58:43.088 MSK [1711] WARNING: setting an MD5-encrypted password
2026-01-13 17:58:43.088 MSK [1711] DETAIL: MD5 password support is deprecated and will be removed in a future release of PostgreSQL.
2026-01-13 17:58:43.088 MSK [1711] HINT: Refer to the PostgreSQL documentation for details about migrating to another password type.
ok
syncing data to disk ... ok

Success. You can now start the database server using:

```

- Using `pg_createcluster`

```

# Create a cluster for specific PostgreSQL version
pg_createcluster 18 mycluster

```

```

postgres@998-986-02:~/18/data$ pg_createcluster 18 mycluster
Creating new PostgreSQL cluster 18/mycluster ...
/usr/lib/postgresql/18/bin/initdb -D /var/lib/postgresql/18/mycluster --auth-local peer --auth-host scram-sha-256 --no-instructions
The files belonging to this database system will be owned by user "postgres".
This user must also own the server process.

The database cluster will be initialized with locale "C.UTF-8".
The default database encoding has accordingly been set to "UTF8".
The default text search configuration will be set to "english".

Data page checksums are enabled.

fixing permissions on existing directory /var/lib/postgresql/18/mycluster ... ok
creating subdirectories ... ok
selecting dynamic shared memory implementation ... posix
selecting default "max_connections" ... 100
selecting default "shared_buffers" ... 128MB
selecting default time zone ... Europe/Volgograd
creating configuration files ... ok
running bootstrap script ... ok
performing post-bootstrap initialization ... ok
syncing data to disk ... ok
Warning: systemd does not know about the new cluster yet. Operations like "service postgresql start" will not handle it. To fix, run:
  sudo systemctl daemon-reload
Ver Cluster  Port Status Owner  Data directory  Log file
18 mycluster 5433 down postgres /var/lib/postgresql/18/mycluster /var/log/postgresql/postgresql-18-mycluster.log
postgres@998-986-02:~/18/data$ |

```

```
# Create with specific port
pg_createcluster 18 mycluster --port 5433
```

```
postgres@998-986-02:~/18$ pg_createcluster 18 mycluster --datadir=/var/lib/postgresql/18/mycluster
Creating new PostgreSQL cluster 18/mycluster ...
/usr/lib/postgresql/18/bin/initdb -D /var/lib/postgresql/18/mycluster --auth-local peer --auth-host scram-sha-256 --no-instructions
The files belonging to this database system will be owned by user "postgres".
This user must also own the server process.

The database cluster will be initialized with locale "C.UTF-8".
The default database encoding has accordingly been set to "UTF8".
The default text search configuration will be set to "english".

Data page checksums are enabled.

fixing permissions on existing directory /var/lib/postgresql/18/mycluster ... ok
creating subdirectories ... ok
selecting dynamic shared memory implementation ... posix
selecting default "max_connections" ... 100
selecting default "shared_buffers" ... 128MB
selecting default time zone ... Europe/Volgograd
creating configuration files ... ok
running bootstrap script ... ok
performing post-bootstrap initialization ... ok
syncing data to disk ... ok
Warning: systemd does not know about the new cluster yet. Operations like "service postgresql start" will not handle it. To fix, run:
  sudo systemctl daemon-reload
Ver Cluster  Port Status Owner    Data directory                   Log file
18  mycluster 5433 down  postgres /var/lib/postgresql/18/mycluster /var/log/postgresql/postgresql-18-mycluster.log
postgres@998-986-02:~/18$
```

- Using pg_ctl

```
# Create and start in one command
pg_ctl init -D /var/lib/postgresql/18/data
```

```
postgres@998-986-02:~/18$ pg_ctl init -D /var/lib/postgresql/18/data
The files belonging to this database system will be owned by user "postgres".
This user must also own the server process.

The database cluster will be initialized with locale "C.UTF-8".
The default database encoding has accordingly been set to "UTF8".
The default text search configuration will be set to "english".

Data page checksums are enabled.

fixing permissions on existing directory /var/lib/postgresql/18/data ... ok
creating subdirectories ... ok
selecting dynamic shared memory implementation ... posix
selecting default "max_connections" ... 100
selecting default "shared_buffers" ... 128MB
selecting default time zone ... Europe/Volgograd
creating configuration files ... ok
running bootstrap script ... ok
performing post-bootstrap initialization ... ok
syncing data to disk ... ok

initdb: warning: enabling "trust" authentication for local connections
initdb: hint: You can change this by editing pg_hba.conf or using the option -A, or --auth-local and --auth-host, the next time you run initdb.

Success. You can now start the database server using:

    /usr/lib/postgresql/18/bin/pg_ctl -D /var/lib/postgresql/18/data -l logfile start

postgres@998-986-02:~/18$
```

- Creating Multiple Clusters

```
# Create development cluster
```

```
pg_createcluster 18 dev --port 5432
```

```
postgres@998-986-02:~$ pg_createcluster 18 dev --port 5432
Creating new PostgreSQL cluster 18/dev ...
/usr/lib/postgresql/18/bin/initdb -D /var/lib/postgresql/18/dev --auth-local peer --auth-host scram-sha-256 --no-instructions
The files belonging to this database system will be owned by user "postgres".
This user must also own the server process.

The database cluster will be initialized with locale "C.UTF-8".
The default database encoding has accordingly been set to "UTF8".
The default text search configuration will be set to "english".

Data page checksums are enabled.

fixing permissions on existing directory /var/lib/postgresql/18/dev ... ok
creating subdirectories ... ok
selecting dynamic shared memory implementation ... posix
selecting default "max_connections" ... 100
selecting default "shared_buffers" ... 128MB
selecting default time zone ... Europe/Volgograd
creating configuration files ... ok
running bootstrap script ... ok
performing post-bootstrap initialization ... ok
syncing data to disk ... ok
Warning: systemd does not know about the new cluster yet. Operations like "service postgresql start" will not handle it. To fix, run:
  sudo systemctl daemon-reload
Ver Cluster Port Status Owner    Data directory          Log file
18 dev      5432 down   postgres /var/lib/postgresql/18/dev /var/log/postgresql/postgresql-18-dev.log
postgres@998-986-02:~$
```

```
# Create testing cluster
```

```
pg_createcluster 18 test --port 5433
```

```
postgres@998-986-02:~$ pg_createcluster 18 test --port 5433
Creating new PostgreSQL cluster 18/test ...
/usr/lib/postgresql/18/bin/initdb -D /var/lib/postgresql/18/test --auth-local peer --auth-host scram-sha-256 --no-instructions
The files belonging to this database system will be owned by user "postgres".
This user must also own the server process.

The database cluster will be initialized with locale "C.UTF-8".
The default database encoding has accordingly been set to "UTF8".
The default text search configuration will be set to "english".

Data page checksums are enabled.

fixing permissions on existing directory /var/lib/postgresql/18/test ... ok
creating subdirectories ... ok
selecting dynamic shared memory implementation ... posix
selecting default "max_connections" ... 100
selecting default "shared_buffers" ... 128MB
selecting default time zone ... Europe/Volgograd
creating configuration files ... ok
running bootstrap script ... ok
performing post-bootstrap initialization ... ok
syncing data to disk ... ok
Warning: systemd does not know about the new cluster yet. Operations like "service postgresql start" will not handle it. To fix, run:
  sudo systemctl daemon-reload
Ver Cluster Port Status Owner    Data directory          Log file
18 test     5433 down   postgres /var/lib/postgresql/18/test /var/log/postgresql/postgresql-18-test.log
postgres@998-986-02:~$
```

```
# Create production cluster
```

```
pg_createcluster 18 prod --port 5434
```

```
postgres@998-986-02:~$ pg_createcluster 18 prod --port 5434
Creating new PostgreSQL cluster 18/prod ...
/usr/lib/postgresql/18/bin/initdb -D /var/lib/postgresql/18/prod --auth-local peer --auth-host scram-sha-256 --no-instructions
The files belonging to this database system will be owned by user "postgres".
This user must also own the server process.

The database cluster will be initialized with locale "C.UTF-8".
The default database encoding has accordingly been set to "UTF8".
The default text search configuration will be set to "english".

Data page checksums are enabled.

fixing permissions on existing directory /var/lib/postgresql/18/prod ... ok
creating subdirectories ... ok
selecting dynamic shared memory implementation ... posix
selecting default "max_connections" ... 100
selecting default "shared_buffers" ... 128MB
selecting default time zone ... Europe/Volgograd
creating configuration files ... ok
running bootstrap script ... ok
performing post-bootstrap initialization ... ok
syncing data to disk ... ok
Warning: systemd does not know about the new cluster yet. Operations like "service postgresql start" will not handle it. To fix, run:
  sudo systemctl daemon-reload
Ver Cluster Port Status Owner    Data directory          Log file
18  prod    5434  down   postgres /var/lib/postgresql/18/prod /var/log/postgresql/postgresql-18-prod.log
postgres@998-986-02:~$
```

- Verifying the Cluster

```
# Check cluster structure
```

```
ls -la /var/lib/postgresql/18/data/
```

```
postgres@998-986-02:~$ ls -la /var/lib/postgresql/18/data/
total 128
drwx----- 19 postgres postgres 4096 Jan 13 18:07 .
drwxr-xr-x  8 postgres postgres 4096 Jan 13 21:54 ..
-rw-----  1 postgres postgres   3 Jan 13 18:07 PG_VERSION
drwx-----  5 postgres postgres 4096 Jan 13 18:07 base
drwx-----  2 postgres postgres 4096 Jan 13 18:07 global
drwx-----  2 postgres postgres 4096 Jan 13 18:07 pg_commit_ts
drwx-----  2 postgres postgres 4096 Jan 13 18:07 pg_dynshmem
-rw-----  1 postgres postgres 5721 Jan 13 18:07 pg_hba.conf
-rw-----  1 postgres postgres 2681 Jan 13 18:07 pg_ident.conf
drwx-----  4 postgres postgres 4096 Jan 13 18:07 pg_logical
drwx-----  4 postgres postgres 4096 Jan 13 18:07 pg_multixact
drwx-----  2 postgres postgres 4096 Jan 13 18:07 pg_notify
drwx-----  2 postgres postgres 4096 Jan 13 18:07 pg_replslot
drwx-----  2 postgres postgres 4096 Jan 13 18:07 pg_serial
drwx-----  2 postgres postgres 4096 Jan 13 18:07 pg_snapshots
drwx-----  2 postgres postgres 4096 Jan 13 18:07 pg_stat
drwx-----  2 postgres postgres 4096 Jan 13 18:07 pg_stat_tmp
drwx-----  2 postgres postgres 4096 Jan 13 18:07 pg_subtrans
drwx-----  2 postgres postgres 4096 Jan 13 18:07 pg_tblspc
drwx-----  2 postgres postgres 4096 Jan 13 18:07 pg_twophase
drwx-----  4 postgres postgres 4096 Jan 13 18:07 pg_wal
drwx-----  2 postgres postgres 4096 Jan 13 18:07 pg_xact
-rw-----  1 postgres postgres   88 Jan 13 18:07 postgresql.auto.conf
-rw-----  1 postgres postgres 32374 Jan 13 18:07 postgresql.conf
postgres@998-986-02:~$
```

- DELETE a Cluster

Using `pg_dropcluster`.

```
# List clusters first to identify
pg_lsclusters
```

```
postgres@998-986-02:~$ pg_lsclusters
Ver Cluster Port Status Owner  Data directory          Log file
18 dev      5432 down  postgres /var/lib/postgresql/18/dev /var/log/postgresql/postgresql-18-dev.log
18 main     5432 online postgres /var/lib/postgresql/18/main log/postgresql-%Y-%m-%d_%H%M%S.csv.csv
18 mycluster 5433 online postgres /var/lib/postgresql/18/mycluster /var/log/postgresql/postgresql-18-mycluster.log
18 prod     5434 down  postgres /var/lib/postgresql/18/prod /var/log/postgresql/postgresql-18-prod.log
18 test     5433 down  postgres /var/lib/postgresql/18/test /var/log/postgresql/postgresql-18-test.log
postgres@998-986-02:~$
```

```
# Stop the cluster
pg_ctlcluster 18 mycluster stop
```

```
postgres@998-986-02:~$ pg_lsclusters
Ver Cluster Port Status Owner  Data directory          Log file
18 dev      5432 down  postgres /var/lib/postgresql/18/dev /var/log/postgresql/postgresql-18-dev.log
18 main     5432 online postgres /var/lib/postgresql/18/main log/postgresql-%Y-%m-%d_%H%M%S.csv.csv
18 mycluster 5433 down  postgres /var/lib/postgresql/18/mycluster /var/log/postgresql/postgresql-18-mycluster.log
18 prod     5434 down  postgres /var/lib/postgresql/18/prod /var/log/postgresql/postgresql-18-prod.log
18 test     5433 down  postgres /var/lib/postgresql/18/test /var/log/postgresql/postgresql-18-test.log
postgres@998-986-02:~$
```

```
# Drop the cluster
pg_dropcluster 18 mycluster -stop
```

```
postgres@998-986-02:~$ pg_lsclusters
Ver Cluster Port Status Owner  Data directory          Log file
18 dev      5432 down  postgres /var/lib/postgresql/18/dev /var/log/postgresql/postgresql-18-dev.log
18 main     5432 online postgres /var/lib/postgresql/18/main log/postgresql-%Y-%m-%d_%H%M%S.csv.csv
18 prod     5434 down  postgres /var/lib/postgresql/18/prod /var/log/postgresql/postgresql-18-prod.log
18 test     5433 down  postgres /var/lib/postgresql/18/test /var/log/postgresql/postgresql-18-test.log
postgres@998-986-02:~$ pg_dropcluster 18 test --stop
Warning: systemd was not informed about the removed cluster yet. Operations like "service postgresql start" might fail. To fix, run:
  sudo systemctl daemon-reload
postgres@998-986-02:~$ systemctl daemon-reload
==== AUTHENTICATING FOR org.freedesktop.systemd1.reload-daemon ====
Authentication is required to reload the systemd state.
Authenticating as: ,, (steve)
Password:
==== AUTHENTICATION COMPLETE ====
postgres@998-986-02:~$ pg_lsclusters
Ver Cluster Port Status Owner  Data directory          Log file
18 dev      5432 down  postgres /var/lib/postgresql/18/dev /var/log/postgresql/postgresql-18-dev.log
18 main     5432 online postgres /var/lib/postgresql/18/main log/postgresql-%Y-%m-%d_%H%M%S.csv.csv
18 prod     5434 down  postgres /var/lib/postgresql/18/prod /var/log/postgresql/postgresql-18-prod.log
postgres@998-986-02:~$
```